

# Sanchit Acharya

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linkedin.com/in/acharyasanchit | github.com/sanchitacharya | Portfolio

## Education

### University of Hertfordshire

BEng (Hons) Robotics and Artificial Intelligence

Hatfield, UK

Sept 2025 – Present

First-year GPA: **4.16 / 4.5**. Seeking a placement year for 2027–28.

Relevant modules: Maths for Robotics and AI (96%), Introduction to Robotics (85%), Robot Design and Build (85%), Digital Electronics (84%, including PCB design and logic circuits), Programming (83%).

## Experience

### Software Lead

UH Racing Autonomous (UHRA)

Feb 2026 – Present

Hatfield, UK

Software Lead since Sept 2026; software team member from Feb 2026.

- Leading the software team towards Formula Student AI 2027 (July 2027).
- Built acceleration planning and control; UHRA scored its first ever acceleration point.
- Built a skidpad mission in about six hours in simulation.

### Robotics Engineer

STRIDE, University of Hertfordshire

Nov 2025 – Present

Hatfield, UK

- Multi-object tracking of robots and the ball, in simulation and on an OAK-D camera.
- Odometry-based localization with visual landmarks for ROBOTIS OP3 humanoids in Webots.

### SPECS Proctor (Engineering)

University of Hertfordshire

May 2026 – Present

Hatfield, UK

- Assist academics in engineering labs and lectures, help students with technical work, and support Open Days.

### R&D Intern

Robotics Association of Nepal

Jun 2024 – Feb 2025

Lalitpur, Nepal

- Robotics and applied AI projects: drone-based afforestation, banana fibre production, and object detection.
- Helped organise the International AI & Robotics Conference; ran workshops in drone simulation, 3D printing, and robotics, and mentored students in Girls in Robotics & AI.

### Team Leader (Shift Manager)

Co-op Food

Oct 2025 – Present

Hatfield, UK

- Run shifts alongside full-time study: lead the team, allocate tasks, and handle customer issues.

## Projects

### Formula Student AI

UHRA

2026

write-up

- Planning and control for a driverless car: first acceleration point, skidpad lessons, lidar + IMU localization plan.

### Visual Localisation for Humanoid Robots

STRIDE

2025 – Present

write-up

- Phase 1: ground-truth tracking system for robots and the ball, with stable IDs in field coordinates.
- Phase 2: the robot localises itself from goalposts and IMU heading, and remembers the ball's last position.

### PID Line Follower

University coursework

2025

code

- PID line-following Thymio robot with obstacle detection and line recovery. Tuned over 15+ runs: 20% less motion jitter, 30% smoother cornering.

### DIY Quadcopter Drone

Robotics Association of Nepal

2024 – 2025

- Built, configured, and flew a quadcopter; improved flight stability by 30% through tuning over 20+ test flights.

## Skills

**Programming & tools:** Python, C++, Linux, Git, Docker

**Robotics:** ROS 2 (Humble, Jazzy), Gazebo, Webots, RViz, kinematics

**Autonomy:** planning and control, localization, Kalman filtering, multi-object tracking

**CAD & fabrication:** Fusion 360, 3D printing, DesignSpark PCB; built a 3D-printed RC car (servo, DC motor)

**Drones:** building, configuring, and flying DIY drones (QGroundControl, Mission Planner)